CO3 – Home Assignment

1. A pharmaceutical company claims that the average recovery time after using its medicine is less than 8 days. A random sample of 36 patients has a mean recovery time of 7.3 days. The population standard deviation is 1.8 days. Formulate the appropriate null and alternative hypotheses, identify whether the test is one-sided or two-sided, compute the Z-test statistic, and conclude at the 5% significance level.
2. A manufacturer claims that the mean diameter of machine parts is 20 mm. A quality engineer suspects the mean has changed. A sample of 49 parts has a mean diameter of 19.4 mm with a population standard deviation of 2.1 mm. Perform the appropriate hypothesis test at $\alpha = 0.01$ and justify whether a one-sided or two-sided test is appropriate.
3. A sample of 25 observations yields a test statistic of $t = 2.36$ while testing a population mean. The significance level is 5%. Determine the degrees of freedom, obtain the p-value (or compare with the critical value), state the decision, and explain whether a Type I error is possible if the null hypothesis is rejected.
4. Two teaching methods are compared. Method A: n=15, mean=74.2, s=6.8. Method B: n=14, mean=69.5, s=5.9. Assuming equal variances, perform an independent two-sample t-test at the 5% significance level to determine whether the teaching methods differ significantly.
5. The weights (kg) of 10 athletes are measured before and after an 8-week fitness program. Before: 82, 79, 85, 88, 76, 81, 90, 84, 77, 83. After: 79, 77, 82, 85, 74, 79, 87, 81, 75, 80. Perform a paired t-test to determine whether the program significantly reduces body weight at $\alpha = 0.05$.
6. A die is rolled 180 times and the observed frequencies are 24, 29, 31, 33, 30, and 33. Using the Chi-square goodness-of-fit test, determine whether the die is fair at the 5% significance level.
7. A survey records the preferred mode of online shopping among three age groups. The observed data are:
   18–30: Mobile=48, Laptop=32, Tablet=20
   31–50: Mobile=36, Laptop=44, Tablet=20
   Above 50: Mobile=22, Laptop=30, Tablet=28.
   Perform a Chi-square test of independence to determine whether shopping preference depends on age group.
8. Three fertilizers are tested on crop yield with 10 plots in each group. Identify the suitable statistical test and state the null and alternative hypotheses for the study.
9. A survey records customer satisfaction as Satisfied and Not Satisfied across three different branches of a bank. Identify the appropriate Chi-square test and write the hypotheses to be tested.
10. A researcher wants to determine whether the average monthly electricity consumption differs among four residential areas. A random sample of households from each area produced the following electricity consumption (in kWh):

Area A	Area B	Area C	Area D
210	235	260	285
220	240	255	290
215	245	265	295
225	250	270	300
230	255	275	305

    Using an appropriate statistical test, determine whether there is a significant difference in the mean electricity consumption among the four residential areas at the 5% significance level. Clearly state the hypotheses, perform all necessary calculations, compute the ANOVA table, determine the test statistic, compare it with the critical value (or use the p-value), and draw an appropriate conclusion.